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Agronomic traits of soybean cultivars released in different decades after grafting record-yield cultivar as rootstock

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Abstract

Breeders have seldom considered the selection for root traits during the genetic improvement in soybean. It is hypothesized that grain yield would be increased by the root function improvement, especially for the current cultivars. The objective of this grafting experiment was to determine the effect of record-yield cultivars L14 or Z35 as rootstocks on agronomic traits of cultivars released in different decades. A total of 11 cultivars, released in different decades, were used to graft onto L14 or Z35 rootstocks. The agronomic traits were measured in the pot-culture experiments. Grafting cultivars released in different decades onto L14 or Z35 rootstocks resulted in higher yield, 100-seed mass and apparent harvest index as compared with those of non-grafted or self-grafted plants. Grain yield gain of cultivars grafted onto record-yield cultivar rootstocks was 0.40 g/plant/year from 1966 to 2006, which was larger than that of non-grafts and self-grafts (0.27 g/plant/year). The yield of current cultivars should increase more if their root functions were improved.

Key words: grain yield — genetic·gain — root function — soybean (*Glycine max*)

Grain yield and some agronomic traits of soybean (*Glycine max*) have been significantly improved during several decades by breeding. In the north central region of the USA, Specht and Williams (1984) reported that the yield improvement from 10 to 20 kg/ha/year was achieved from cultivars released in 1902–1977 in maturity groups 00 through IV. Rogers et al. (2015) reported an yield gain of 16.8 kg/ha/year from cultivars released in 1928–2008 in maturity groups IV through V. In Northeast China, Jin et al. (2010) found a 0.58% yield improvement in 45 cultivars of maturity groups 00 and 0 from 1950 to 2006, which had an association with the increase in seed number per plant, photosynthetic rate, dry weight, harvest index, and the decrease in leaf area index. Xie et al. (2010) used the older cultivars and their modern counterparts derived from breeding programmes in Liaoning (China) and Ohio (USA) to indicate that the improvement in lodging resistance was the most important contributor to yield gain. Recently, a larger increase of 23.4 kg/ha/year between 1924 and 2011 in the United States has been reported (USDA-ERS 2011). But this current rate of yield gain is insufficient to meet the target of doubling crop yields by 2050 (Ray et al. 2013).

There were several breeding strategies for obtaining high grain yield, such as the genetic improvement for pods per plant, seeds per pod and seed mass (Cui and Yu 2005). Although the yield components have largely improved by breeding, many factors impacted the expression of yield potential. Some studies showed that the modern soybean cultivars could be yielded more than older cultivars, especially in high-yielding environments (Rincker et al. 2014), in non-limiting supply of nitrogen levels (Wilson et al. 2013), in the higher plant populations (Suhre et al. 2014), or with earlier planting date (Rowntree et al. 2013; 2014). Apart from the factors of environment or agronomic practices, however, it also seems that the root functions *per se* of cultivars may be the key limited factors for yield gain. Roots mediate uptakes of water and ions, and respond to external signals, which had significant relationships with the agronomic and yield traits (Pantalone et al. 1996, Cui et al. 2016). But the breeders have seldom considered selection for root traits during the improvement in soybean on account of the difficulty in observing and measuring roots *in situ*. Thus, it is hypothesized that grain yield would be increased by the root function improvement, especially for the recently released cultivars.

Grafting to combine scion and rootstock of differing phenotypes has been used in soybean research (Cardwell and Polson 1972, Pantalone et al. 1999, Ookawa et al. 2001). The grafting could be helpful to test the previous hypothesis by changing the root function while ensuring a consistent shoot genotype. Cultivars ‘Liaodou 14’ (L14) and ‘Zhonghuang 35’ (Z35) had produced the yield records of 4908 and 6089 kg/ha in production practice, respectively (Song et al. 2001, Jin and Wang 2014), are therefore known as the record-yield cultivars in China. They had higher yield potential than common cultivars (Ao et al. 2013), which was attributed to strong root physiological activity (Zhang et al. 2013). In this grafting experiment, the record-yield cultivars were used as the rootstocks to increase the root functions of cultivars released in different decades.

Moreover, in order to eliminate the interference of complex pedigrees among cultivars, a set of somewhat related cultivars, released in different decades by Liaoning, China, and OH, USA, and their the common ancestors, were used to graft onto the record-yield cultivar rootstocks. The objective of this study was to determine (1) the effect of root function improvement on the agronomic traits and (2) the yield gains of cultivars released in different decades without the restriction of their root functions.

Materials and Methods

Plant materials and grafting procedure: A total of 11 cultivars of maturity group III, released in different decades by Liaoning (38.55°~42.32°N), China, and Ohio (38.45°~41.22°N), USA, and their common ancestors Amsoy and Williams (*i.e.*, excellent ancestors in USA and China, Gizlice et al. 1994, Cui et al. 2002), were collected (Fig. 1). According to the release years and origins, these cultivars were classified into five groups, including common parents, Liaoning middle, Ohio middle, Liaoning current and Ohio current cultivars (Table 1). L14

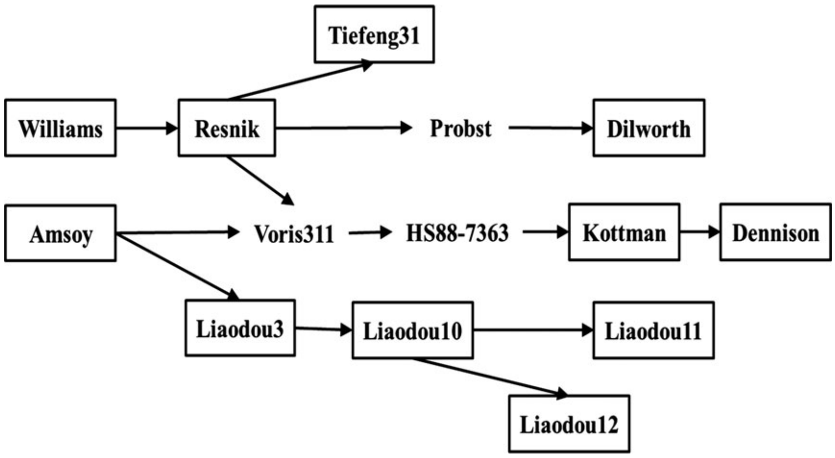

Fig. 1: Genetic relationship of soybean cultivars tested in this study (boxed)

Table 1: Cultivar name, year of release and origin of soybean cultivars tested in this study

Group	Cultivar	Growth habit	Year of release	Pedigree	Origin
Common parents	Amsoy	Indeterminate	1966	Adams × Harosoy	Iowa State Univ., USA
	Williams	Indeterminate	1971	Wayne × L57-0034	USDA, Univ. of Illinois., USA
Liaoning middle	Liaodou3	Semi-determinate	1983	Tiefeng18 × Amsoy	Liaoning Academy of Agric.Sci., CN
	Liaodou10	Semi-determinate	1991	Liaodou3 × Liao82-5185	Liaoning Academy of Agric.Sci., CN
Ohio middle	Resnik	Indeterminate	1987	A3127 × L24	Ohio State Univ., USA
	Kottman	Indeterminate	1991	HS88-7363 × HS88-4988	Ohio State Univ., USA
Liaoning current	Liaodou11	Semi-determinate	1996	Liao84063 × Liaodou3	Liaoning Academy of Agric.Sci.,CN
	Liaodou12	Semi-determinate	2001	Liao85094 × Liaodou 10	Liaoning Academy of Agric.Sci., CN
	Tiefeng31	Semi-determinate	2001	Xin3511 × Resnik	Tieling Academy of Agric.Sci., CN
Ohio current	Dilworth	Indeterminate	2002	Chapman × Probst	Ohio State Univ., USA
	Dennison	Indeterminate	2006	Athow × HS94-4533	Ohio State Univ., USA
Record-yield	Liaodou14	Semi-determinate	2003	Liaodou10 × Mecury	Liaoning Academy of Agric.Sci., CN
	Zhonghuang35	Determinate	2006	(PI486355 × Zheng8431) × Zheng6062	Chinese Academy of Agric.Sci., CN

(Liaodou 10 × Mecury) and Z35 [(PI 486355 × Zheng 8431) × Zheng 6062], maturity group III cultivars, were used to represent record-yield soybean cultivars.

Scions of the 11 cultivars released in different decades were grafted onto the rootstocks of L14 (D/L14) and also onto the rootstocks of Z35 (D/Z35), and their means were represented as D/R. The self-grafted and non-grafted plants of 11 cultivars were used as controls, and their means were represented as D/D. Grafting of soybean seedlings was performed as described by Pantalone et al. (1999), with the following additional modifications. Seedlings were planted in soil in 12-cm-diameter plastic pots with soil on 7 May 2014 and 4 May 2015. Immediately after grafting, the grafted and non-grafted plants were placed in a glasshouse with ~70% shade and 23°C~28°C temperature for 2 weeks. Humidifiers were used to keep glasshouse humidity above 90%.

Pot-culture experiments: To simulate field conditions, the pot-culture experiments were carried out under open field conditions in 2014 and 2015 at Shenyang Agricultural University (41°82′N, 123°57′E), Liaoning

Table 2: Analysis of variance (mean square and significance) of agronomic traits evaluated in 2014 and 2015

	Source of variations							
Trait (/plant)	Year (Y)	Genotype (G)	G × Y	Treatment (T)	T × Y	T × G	T × G × Y	Error
Plant height (cm)	894.68 ns	3121.68**	614.63***	31.37 ns	5.11 ns	83.01 ns	89.80***	30.40
Internode length (cm)	6.40 ns	8.11***	2.24*	2.45 ns	0.48 ns	1.59 ns	1.14 ns	0.15
Plant weight balance (cm)	52.74 ns	509.46**	80.32**	27.93 ns	38.32 ns	43.92 ns	26.99*	15.26
Stem diameter (mm)	123.48*	10.57*	2.49 ns	1.92 ns	4.20 ns	3.35 ns	2.11**	1.07
Height of lowest pod (cm)	3637.88**	114.61 ns	52.50*	9.01 ns	73.01*	19.48 ns	21.06 ns	17.98
Branch	20.19 ns	7.05 ns	4.25 ns	1.67 ns	4.90 ns	3.16 ns	4.15***	2.18
Pod number	913.19 ns	2311.46**	440.39*	855.23 ns	307.88 ns	143.02 ns	153.88 ns	130.31
Seed number	73.19 ns	16 497.21**	2274.30***	3816.15 ns	1147.93*	354.63 ns	303.88**	157.91
100-seed mass (g)	136.37**	179.28***	12.03***	56.65***	0.05 ns	4.46*	2.41***	0.82
Biomass (g)	576.55 ns	788.78***	90.07**	289.41 ns	100.60*	24.15 ns	29.16 ns	23.50
Apparent harvest index	0.013 ns	0.036***	0.003*	0.028*	0.002 ns	0.001 ns	0.001 ns	0.001
Yield (g)	308.04*	457.42***	27.43***	309.88*	16.64*	10.19*	4.88***	2.06

*, **, ***Significant at the 0.05, 0.01 and 0.001 probability levels, respectively.
ns, Not significant.

Province, China. After seedling recovering, the survival grafts (survival rate ≥ 90%) and non-grafts were transplanted into the pots (25 × 30 × 25 cm, 12.5 kg soil).

The soil type was brown, and the chemical characteristics were as follows: soil organic matter of 16.87 g/kg, total nitrogen of 0.79 g/kg, available nitrogen of 0.07 g/kg, available phosphorus of 0.02 g/kg, available potassium of 0.14 g/kg and pH of 7.33 in distilled water (1 : 5 v/v). The moisture content in the soil was maintained at ~70% of the field water-holding capacity using drip irrigation. The experiments were conducted without insect and disease stress. There was a randomized complete block design with three replications per treatment, and each pot contained two plants of the same treatment and was considered as one experimental unit.

Agronomic traits measurement: At maturity, two plants of each pot were sampled for the measurement of agronomic traits. Each sampled plant was measured for plant height from the node of cotyledon to the tip of stem, and plant weight balance for the node of cotyledon to the centre of gravity of plant, the mean internode length of the main stem, the height of lowest pod, the numbers of branches, the numbers of pods, the numbers of seeds and seed weight per plant were measured. Seed yield was divided by the total plant weight (stem weight plus pod weight) to determine the apparent harvest index.

Statistical analysis: The experimental data were subjected to an analysis of variance (ANOVA) in a general linear model with 2 years, four grafting treatments and 11 genotypes by SPSS-17.0 (*SPSS Inc*., Chicago, USA). The grafting treatment, genotype and grafting treatment × genotype were regarded as fixed effects, while years, along with the grafting treatment × year, genotype× year, and grafting treatment × genotype × year were regarded as a random factor. Means were subjected to the least significant difference (*LSD*) test at the $P < 0.05$ level. Pearson's product–moment correlations between traits and year of release were determined. Traits were regressed over year of release under the different grafting treatments. The difference of regression coefficients was compared by covariance analysis of general linear model with the grafting treatment as the fixed effects, and the year of release as concomitant variable.

Table 3: Agronomic traits evaluated for four grafting treatments

	Grafting treatment			
Trait (/plant)	Non-graft	Self-graft	D/L14[1]	D/Z35[2]
Plant height (cm)	92.55 a[3]	91.36 a	92.97 a	92.06 a
Internode length (cm)	4.61 a	4.57 a	4.50 a	4.50 a
Plant weight balance (cm)	44.55 a	42.97 a	43.58 a	43.61 a
Stem diameter (mm)	7.59 a	7.26 a	7.64 a	7.56 a
Height of lowest pod (cm)	18.14 a	17.38 a	17.35 a	17.74 a
Branch	5.14 a	4.92 a	5.20 a	4.88 a
Pod number	74.59 a	73.67 a	81.47 a	78.41 a
Seed number	159.27 a	157.56 a	174.09 a	160.33 a
100-seed mass (g)	16.68 b	16.53 b	18.30 a	18.11 a
Biomass (g)	56.90 a	55.23 a	60.25 a	57.16 a
Apparent harvest index	0.46 b	0.47 b	0.50 a	0.50 a
Yield (g)	26.22 c	25.90 c	30.52 a	28.57 b

[1]D/L14, the cultivars released in different decades grafted onto the L14 rootstocks.
[2]D/Z35, the cultivars released in different decades grafted onto the Z35 rootstocks.
[3]Within a row, means followed by the same letter did not differ significantly at 5% levels among different grafting treatments.

Results

Effect of grafting treatments

Table 2 summarizes the mean square of each factor (four grafting treatments, 11 genotypes and 2 years) and their interactions for agronomic traits. There was non-significant effect of grafting treatment on some morphological and yield traits, specifically plant height, internode length, plant weight balance, stem diameter, height of the lowest pod, branch, pod numbers, seed numbers and biomass. But some yield traits including 100-seed mass, apparent harvest index and yield per plant showed a significant effect of grafting treatment. There was no difference of 100-seed mass, apparent harvest index and yield between non-grafts and self-grafts, but the cultivars grafted onto L14 or Z35 rootstocks averaged a significant increase in 100-seed mass, apparent harvest index and yield, respectively, as compared with non-grafts or self-grafts (Table 3).

Genetic gain of cultivars with different grafting treatment

All traits showed significant variation among the cultivars (G) with the exception of pod height and branch number (Table 2). The correlation and regression coefficients of traits on year of cultivar release were shown in Table 4. The plant height,

Table 4: Pearson's correlation coefficients and regression coefficient of traits on year of cultivars release

	Correlation coefficient		Regression coefficient		
Trait (/plant)	D/D[1]	D/R[2]	D/D	D/R	Difference of slopes
Plant height	−0.219*	−0.299***	−0.221	−0.356	ns
Internode length	−0.173*	−0.240**	−0.006	−0.012	ns
Plant weight balance	−0.077 ns	−0.196*	−0.040	−0.103	ns
Stem diameter	0.012 ns	0.068 ns	0.001	0.009	ns
Height of lowest pod	0.085 ns	0.095 ns	0.042	0.048	ns
Branch	0.159 ns	0.135 ns	0.020	0.020	ns
Pod number	0.462***	0.547***	0.537	0.715	ns
Seed number	0.498***	0.501***	1.114	1.345	ns
100-seed mass	0.295***	0.517***	0.063	0.141	**
Biomass	0.585***	0.673***	0.343	0.448	ns
Apparent harvest index	0.498***	0.711***	0.002	0.003	*
Yield	0.864***	0.858***	0.266	0.395	***

[1]D/D, the means of non-grafted and self-grafted of cultivars released in different decades.
[2]D/R, the means of cultivars released in different decades grafted onto L14 and Z35 rootstocks.
*, **, ***Significant at the 0.05, 0.01 and 0.001 probability levels, respectively.
ns, Not significant.

internode length and plant weight balance had negative correlation and regression coefficients on year of cultivars release. The pod numbers, seed numbers, 100-seed mass, biomass, apparent harvest index and yield had positive correlation and regression coefficients on year of cultivar release. Although the regression coefficients of these traits were changed by the grafting treatment, only the yield, 100-seed mass and apparent harvest index were significant. The plants of cultivars grafted onto record-yield cultivar rootstocks showed a greater yield gain than non-grafts and self-grafts (Fig. 2a). For instance, with the non-grafted and self-grafted rootstocks, the yield improvement of 10.36 g/plant had been achieved from cultivars released in 1966 compared with 2006, or the yield was increased with year of cultivar release by 0.27 g/plant/year. While the cultivars were grafted onto record-yield cultivar rootstocks, the yield was increased by 0.40 g/plant/year. Correspondingly, the cultivars grafted onto record-yield cultivar rootstocks also showed a higher annual increase rate of 100-seed mass (0.14 g/year vs. 0.06 g/year, Fig. 2b) and apparent harvest index (0.003 unit/year vs. 0.002 unit/year, Fig. 2c) than non-grafts and self-grafts.

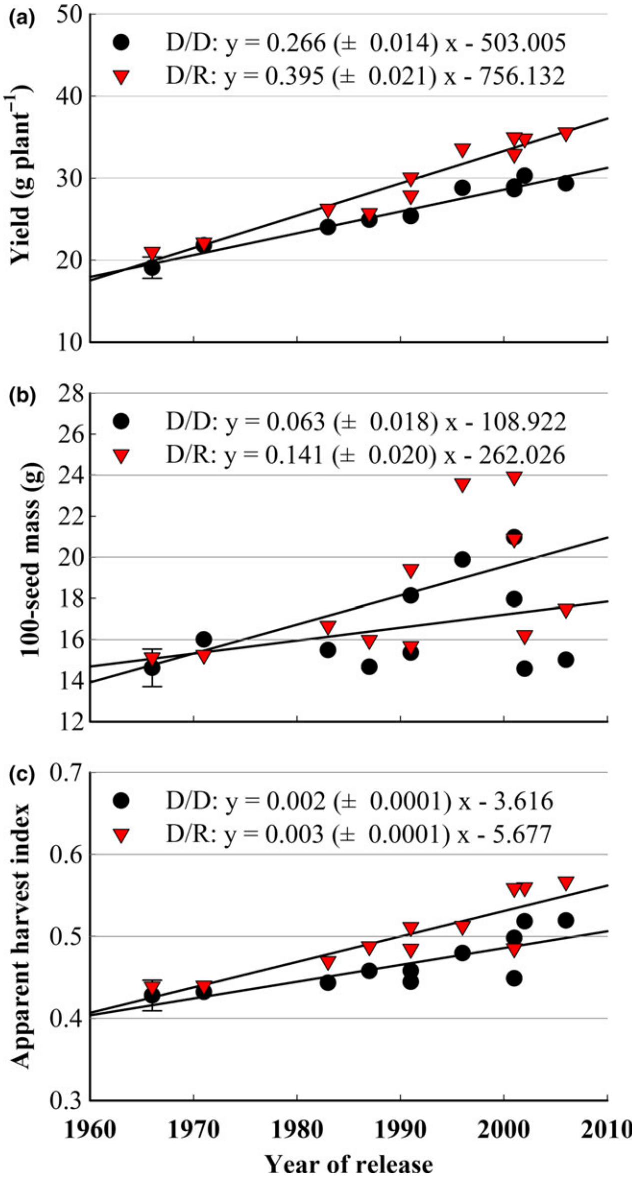

Fig. 2: Regression of (a) yield, (b) 100-seed mass and (c) apparent harvest index on year of cultivar release with different grafting treatments. D/D, the means of non-grafted and self-grafted of cultivars released in different decades; D/R, the means of cultivars released in different decades grafted onto L14 and Z35 rootstocks. The error bar indicates the least significant difference ($P < 0.05$) between treatment × genotype [Colour figure can be viewed at wileyonlinelibrary.com]

Differential response of cultivar groups to record-yield cultivar rootstocks

Interaction between grafting treatment and cultivar was significant for yield (Table 2) due to a differential response of the cultivars to record-yield cultivar rootstocks (Table 5). For instance, the common parents, Liaoning middle and Ohio current cultivars exhibited lower responsiveness to record-yield cultivar rootstocks, with plants grafted onto record-yield cultivar rootstocks having yield that were only 6%, 10% and 11% higher than those of non-grafts and self-grafts, respectively. Conversely, the current cultivars exhibited relatively higher responsiveness to record-yield cultivar rootstocks, with an increase of approximately 18%. There were also different trends of 100-seed mass among the groups of cultivars by the grafting treatment because of the significant interaction between grafting treatment and cultivar. With the rootstocks of record-yield cultivars, the 100-seed mass of common parents was decreased by 1%, while that of Liaoning middle, Ohio middle, and Liaoning current and Ohio current was increased by 7%, 5%, 16% and 14%, respectively, as compared with their non-grafts and self-grafts. Although the interaction between grafting treatment and cultivar for apparent harvest index was non-significant, the trends of apparent harvest index among the groups of cultivars by the grafting treatment were different. With the record-yield cultivar rootstocks, the apparent harvest index of common parents was increased by 2%, while that of Liaoning middle, Ohio middle, and Liaoning

Table 5: Increase percentage of different cultivars for agronomic traits after grafting onto the rootstocks of record-yield cultivars

	Group				
Trait (/plant)	Common parents	Liaoning middle	Ohio middle	Liaoning current	Ohio current
	Increase (%)[1]				
Plant height	1.56	4.67	0.10	1.64	−6.11
Internode length	0.75	−0.71	−4.25	−1.25	−5.04
Plant weight balance	−0.09	1.61	0.83	2.44	−8.54
Stem diameter	−2.86	15.40	−9.47	13.48	−8.31
Height of lowest pod	−4.93	−0.23	4.87	−0.15	−5.97
Branch	−6.05	29.02	−10.55	−7.31	8.88
Pod number	1.57	4.99	14.44	6.61	10.90
Seed number	5.48	2.80	5.47	6.24	6.93
100-seed mass	−0.82	7.32	5.42	16.34**	13.88*
Biomass	2.86	1.99	1.02	7.18	8.47
Apparent harvest index	2.14	7.48	9.08*	9.13*	8.58*
Yield	5.47	9.63	10.85*	17.52**	17.97**

[1]Increase (%) = (D/R-D/D)/D/D × 100%, D/D, the means of non-grafted and self-grafted of cultivars released in different decades; D/R, the means of cultivars released in different decades grafted onto L14 and Z35 rootstocks.

*, **Significant at the 0.05 and 0.01 probability levels, respectively.

current and Ohio current was increased by 7%, 9%, 9% and 9%, respectively, as compared with their non-grafts and self-grafts.

Discussion

Pantalone et al. (1999) found an increase in seed dry mass via grafting onto the rootstocks of PI416937, which had extensive fibrous-like root system and greater nitrogen fixation. The record-yield cultivars L14 and Z35 had higher root physiological activity during the grain-filling stage than common cultivars, with respect to greater root bleeding sap mass, root activity, nodule weight, nitrogenase activity and amount of nitrogen fixation (Zhang et al. 2013). In this study, plants grafted onto L14 or Z35 rootstocks resulted in a higher grain yield than that of their non-grafts or self-grafts, which was attributed to the increase of 100-seed mass. The betterment of root functions could contribute to the enhanced photosynthesis of soybean (Cui et al. 2016). High photosynthetic capacity would result in greater efficiency at producing carbon resources (Boerma and Ashley 1988, Morrison et al. 1999). Although L14 and Z35 exhibited a stronger capacity of nutrient absorption (Zhao et al. 2014), when they were grafted as rootstocks, they could not result in the increase in pod number and seed number of cultivars released in different decades. In this study, all cultivars released in different decades are with indeterminate and semi-determinate growth habits, which generally had longer flowering periods relative to determinate cultivars (Gai et al. 1984, Foley et al. 1986). These cultivars used in this study perhaps showed less nutritional stress during their longer flowering periods than determinate cultivars, thus showing little response of pod and seed number to root improvement. In addition, L14 or Z35 as rootstocks had no influence on the plant height and internode length, because more nutrients were transported to seeds rather than stem. The record-yield cultivars as rootstocks promoted the transportation and partition of assimilate into seeds, so as to increase the apparent harvest index.

In this study, the plants of cultivars grafted onto record-yield cultivar rootstocks showed a higher yield gain from 1966 to 2006 than the non-grafts and self-grafts. On the one hand, the older cultivars (common parents and middle cultivars) had a slight increase in yield with record-yield cultivar rootstocks indicated that the nutrient requirement for yield formation of older cultivars could be satisfied by absorption of their root *per se* rather than record-yield cultivar rootstocks. Wilson et al. (2013) showed that nitrogen provided by the root system's absorption from soils and biological nitrogen fixation without additional nitrogen nutrition provided almost the entire amount of nitrogen nutrition needed to maximize the yield potential of the older cultivars. The most likely reason was that the weaker sink capacity (seed numbers and seed sizes) of older cultivars could not store more excess products from absorption of record-yield cultivar rootstocks. On the other hand, the current cultivars had higher yield potential, which could be expressed if their root functions were improved. During the genetic improvement, the pod numbers, seed numbers and seed sizes of current cultivars released in Liaoning and Ohio were significantly increased (Xie et al. 2010); however, greater yields may have not been expressed because of suppression of their roots *per se*. Although the root morphological and physiological traits were relevant to the capacity of nutrient absorption, they had been seldom selected in the breeding. The present study suggested that the root functions were the limiting factors to further increase yield for current cultivars.

Conclusions

Modern cultivars have greater yield potential than historical cultivars because of the positive genetic gain. But the modern cultivars need more nutrients and waters from soil to express the yield potential. Increasing fertilizer or irrigation to meet the need of plants is unlikely to be a sustainable solution to improving soybean yields. The record-yield cultivars were used as rootstocks in this grafting experiment to reveal the important role of root function and provide an improvement strategy to promote the exploitation of yield advantage. The root functions improvement could lead to an increase in yield, particularly for the current cultivars with high yield potential. Thus, with the selection for yield traits, breeders should pay more attention to the root functions improvement in order to express the yield potential sufficiently.

Acknowledgements

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